

Panagiotis Aigaias Dimtsoudis

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Education

University of Pennsylvania

M.S.E. in Electrical Engineering

Philadelphia, PA

Expected May 2028

B.S.E. in Electrical Engineering | **GPA: 3.68/4.00**

Expected May 2028

Relevant Coursework: Digital Integrated Circuits and VLSI Fundamentals; Analog Integrated Circuits; Nanoscale Science and Engineering; Power Electronics; Fundamentals of Solid-State Circuits; Embedded Systems Laboratory.

Skills

Programming: C/C++, Python, Java, OCaml, Assembly

Embedded Systems: ATmega328PB, UART, I2C, ADC, PWM, BLE

Hardware: Analog circuit design, PCB layout, soldering, circuit debugging

Test & Characterization: Oscilloscopes, cryogenic probe stations, J-V, C-V, PUND, endurance testing

Design Tools: Altium Designer, LTspice, SolidWorks, Fusion 360

Languages: Greek (Native), English (Fluent), German (Intermediate), Japanese (Intermediate)

Experience

Device Research and Engineering Laboratory

Philadelphia, PA

Research Assistant

May 2026 – Present

- Characterized 6 nm AlN/AlScN/AlN junctions from 14–400 K using J-V, C-V, PUND, breakdown, and endurance tests.
- Developed Python scripts to extract device parameters and compare switching, retention, and endurance.
- Contributed measurements and analysis to an IEDM 2026 submission reporting ON/OFF ratios above 2 across 14–400 K.
- Built 3D CAD models of the device stack and an 8×8 crossbar array to support compute-in-memory design.

ESE 2180: Electronic, Photonic, and Electromechanical Devices

Philadelphia, PA

Teaching Assistant

Aug 2026 – Present

- Develop homework assignments, lab worksheets, and exams for an 80+ student device physics course.
- Set up equipment and teach labs, connecting device physics to hands-on experiments.
- Hold weekly office hours on transistors, LEDs, and photovoltaics; grade work and explain concepts through feedback.

Technology Support Services

Philadelphia, PA

Technology Support Specialist

Jan 2025 – Present

- Resolve student hardware/software issues, document tickets, and coordinate escalations and equipment loans.

Automation Systems Hellas

Thessaloniki, Greece

Computer Engineering Intern

Jun 2024 – Aug 2024

- Developed C and Python utilities for log parsing and configuration checks to support commissioning.
- Updated wiring and I/O documentation, imaged PCs and HMIs, and monitored servers and network devices.

Project Portfolio

Metal Detector PCB

- Designed and routed an analog PCB with ALD1103 MOSFETs, LC oscillators, a mixer, and filters.
- Compared bench measurements with LTspice: fixed oscillator at 50.2 kHz versus 50.27 kHz simulated.
- Assembled and soldered the PCB; used an oscilloscope to debug each stage and verify output to an 8Ω speaker.

Wireless ECG Monitor — Analog Front-End & BLE

- Built an AD623 instrumentation-amplifier front end with virtual-ground biasing and high-pass/low-pass filtering.
- Measured approximately 61 dB maximum gain in stage testing and a 3 dB roll-off near 190.5 Hz.
- Streamed ECG over BLE at ~ 200 Hz; wrote Python scripts for logging, plotting, and R-peak heart-rate estimation.

Guitar Hero Synthesizer

- Wrote bare-metal C for an ATmega328PB synthesizer with PWM audio, ADC whammy input, and an I2C LCD.
- Implemented fret/strum note control, power-chord playback, vibrato, and note decay in a working handheld instrument.
- Resolved speaker-induced shutdowns and audio noise using a 12 V battery and two 5 V regulators.

EcoBeam — Solar-Powered Trail Sensor Network

- Collaborated on an Arduino/XBee trail-node prototype with solar power, an 8×8 LED display, and environmental sensors.
- Implemented information and distress modes with message routing and acknowledgments between radio-linked nodes.
- Developed a test plan for communication range, packet delivery rate, and power draw.